R22

13/11/2025

Code No.: 22ITC10

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. V Sem (Main/Backlog) Examination November 2025

Computer Networks

(Common to CSE, IT, AI&DS)

Time: 3 Hours **Max Marks: 60**

Note: "Answer **ALL** questions from **Part-A** and **Part-B** (including **internal choices**) at one place and in the same order as in the question paper."

Part - A

(5Q X 2M = 10 Marks)

			M	CO	BT
1		Assume 7 devices are arranged in a mesh topology. How many cables are needed and how many ports are needed for each device?	(2)	1	2
2		Discuss about framing in data link layer.	(2)	2	2
3		Describe the Optimality Principle in routing.	(2)	3	2
4		List any two differences between TCP sockets and UDP sockets.	(2)	4	1
5		What is Fully Qualified Domain Name (FQDN) and Partially Qualified Domain Name (PQDN), Give an example of each one.	(2)	5	2

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Draw and explain in detail about the ISO-OSI Reference Model.	(6)	1	3
	(b)	Discuss in detail the of Un-guided transmission media.	(4)	1	2
		(OR)			
7	(a)	What is Computer Network? Explain the applications of Computer Networks.	(4)	1	2
	(b)	Illustrate the concept of packet switching, circuit switching, and virtual circuit switching with suitable diagrams.	(6)	1	2
8	(a)	A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1. What is the actual bit string transmitted? Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?	(5)	2	4
	(b)	Differentiate between collision-free protocols and collision-detection-based protocols with examples.	(5)	2	4
		(OR)			
9	(a)	Demonstrate the functioning of a one-bit sliding window protocol with an example.	(5)	2	3
	(b)	Explain about the Ethernet.	(5)	2	2
10	(a)	A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. a. What is the first address in the block? b. Find the last address for the block c. Find the number of addresses	(5)	3	4

	(b)	What is Count to infinity Problem? Where we will get this problem and what are the solutions to avoid this problem.	(5)	3	2
		(OR)			
11	(a)	State the major difference between distance vector routing and link state routing.	(5)	3	4
	(b)	Write a short note on ICMP Protocols.	(5)	3	2
12	(a)	Discuss the Berkeley Socket API. How do TCP and UDP sockets differ?	(5)	4	2
	(b)	Discuss connection establishment and connection release in TCP with a suitable diagram.	(5)	4	3
		(OR)			
13	(a)	Demonstrate the structure of the TCP segment header with a neat diagram and explain each field.	(5)	4	2
	(b)	What is congestion? Illustrate how to control congestion in a network.	(5)	4	4
14	(a)	Illustrate the Architecture of e-mail.	(5)	5	2
	(b)	Differentiate between streaming stored media and streaming live media with suitable use cases.	(5)	5	4
		(OR)			
15	(a)	Write short notes on IMAP and MIME with an example.	(5)	5	2
	(b)	Compare DNS iterative resolution and recursive resolution with examples.	(5)	5	2

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